Nmap Scanning and Network Traffic Analysis

Topic: Network Traffic Analysis and Nmap.
Task: Perform Nmap scans against Ubuntu and Windows virtual machines

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1. Objective

The objective of this practical is to understand how Nmap identifies live hosts, discovers open ports, detects running services, identifies operating systems, and performs comprehensive network scans.

The practical involves scanning two virtual machines:

- Ubuntu Virtual Machine
- Windows Virtual Machine

The following Nmap scan types are performed:

1. -sn - Ping Scan / Host Discovery
2. -sS - TCP SYN Scan
3. -sV - Service Version Detection
4. -O - Operating System Detection
5. -A - Aggressive Scan

All scan results are saved in both normal and XML formats using -oN and -oX.
Additionally, the TCP SYN scan is captured in Wireshark to understand how port scanning appears in network traffic.

2. Lab Environment

Component	Configuration
Attacker Machine	Kali Linux
Target Machine 1	Ubuntu Linux
Target Machine 2	Windows
Scanning Tool	Nmap
Packet Analysis Tool	Wireshark
Network	Host-only / NAT
Output Formats	Normal (.txt), XML (.xml)

3. Lab Preparation

Step 1: Identify IP Addresses

On Kali Linux, execute:

```
ip addr
```

On Ubuntu, execute:

```
ip addr
```

On Windows, open Command Prompt and execute:

```
ipconfig
```

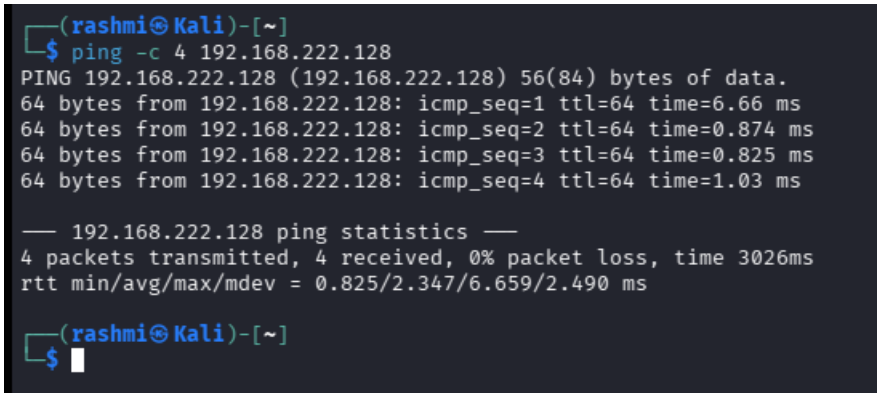
Record the IP addresses of all three machines.

Step 2: Verify Network Connectivity

From Kali Linux, test connectivity with the target machines.

```
ping -c 4 <UBUNTU_IP>
```

ping -c 4 <WINDOWS_IP>



Observation:

- Successful ping responses indicate that the target is reachable through ICMP.
- No response does not necessarily mean the target is offline because firewalls may block ICMP traffic.

Step 3: Install and Verify Nmap

If Nmap is not installed:

sudo apt update

sudo apt install nmap -y

Verify installation:

nmap --version

4. Nmap Scanning Activities

The following scans are performed against both Ubuntu and Windows virtual machines.

4.1. Ping Scan (-sn)

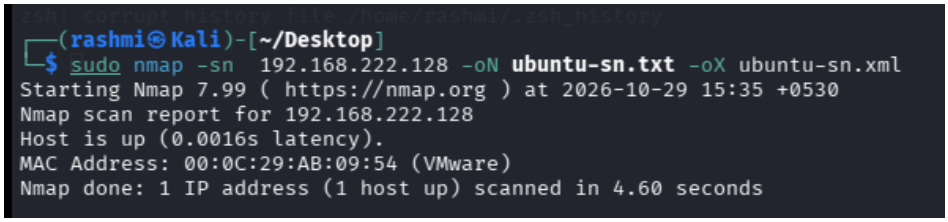
Purpose: The -sn option performs host discovery without performing a port scan.

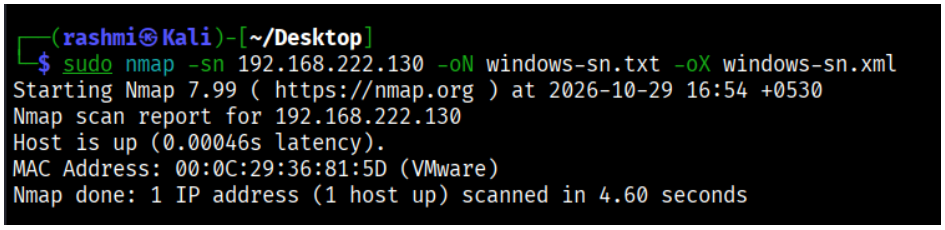
Command for Ubuntu:

sudo nmap -sn <UBUNTU_IP> -oN ubuntu-sn.txt -oX ubuntu-sn.xml

Command for Windows:

sudo nmap -sn <WINDOWS_IP> -oN windows-sn.txt -oX windows-sn.xml





How it works:

- Checks whether the target host is reachable.
- May use ICMP echo requests, TCP probes, or other discovery methods depending on privileges and network configuration.
- Does not perform a port scan.

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Security Relevance: Host discovery is useful for identifying active devices on a network during network reconnaissance.

4.2. TCP SYN Scan (-sS)

Purpose: The -sS option performs a TCP SYN scan, also known as a half-open scan.

Command for Ubuntu:

sudo nmap -sS <UBUNTU_IP> -oN ubuntu-sS.txt -oX ubuntu-sS.xml

Command for Windows:

sudo nmap -sS <WINDOWS_IP> -oN windows-sS.txt -oX windows-sS.xml

```
┌──(rashmi㉿Kali)-[~/Desktop]
└─$ sudo nmap -sS 192.168.222.128 -oN ubuntu-sS.txt -oX ubuntu-sS.xml
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-29 15:35 +0530
Nmap scan report for 192.168.222.128
Host is up (0.00093s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
80/tcp open  http
MAC Address: 00:0C:29:AB:09:54 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 4.90 seconds
```

```
┌──(rashmi㉿Kali)-[~/Desktop]
└─$ sudo nmap -sS 192.168.222.130 -oN windows-sS.txt -oX windows-sS.xml
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-29 16:54 +0530
Nmap scan report for 192.168.222.130
Host is up (0.00076s latency).
All 1000 scanned ports on 192.168.222.130 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:0C:29:36:81:5D (VMware)

Nmap done: 1 IP address (1 host up) scanned in 26.62 seconds
```

How it works:

a. Nmap sends a TCP SYN packet to a target port.
b. If the port is open, the target generally responds with SYN-ACK.
c. Nmap sends RST to terminate the connection instead of completing the TCP handshake.
d. If the port is closed, the target generally responds with RST.
e. Filtered ports may not respond or may return an ICMP error.

Security Relevance: A large number of SYN packets sent to multiple ports within a short period can indicate port scanning or reconnaissance activity.

4.3. Service Version Detection (-sV)

Purpose: The -sV option identifies the services and their versions running on open ports.

Command for Ubuntu:

sudo nmap -sV <UBUNTU_IP> -oN ubuntu-sV.txt -oX ubuntu-sV.xml

Command for Windows:

sudo nmap -sV <WINDOWS_IP> -oN windows-sV.txt -oX windows-sV.xml

```
┌──(rashmi㉿Kali)-[~/Desktop]
└─$ sudo nmap -sV 192.168.222.128 -oN ubuntu-sV.txt -oX ubuntu-sV.xml
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-29 15:36 +0530
Nmap scan report for 192.168.222.128
Host is up (0.0011s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
80/tcp open  http    Apache httpd 2.4.58 ((Ubuntu))
MAC Address: 00:0C:29:AB:09:54 (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 11.37 seconds
```

```
┌──(rashmi㉿Kali)-[~/Desktop]
└─$ sudo nmap -sV 192.168.222.130 -oN windows-sV.txt -oX windows-sV.xml
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-29 16:55 +0530
Nmap scan report for 192.168.222.130
Host is up (0.00053s latency).
All 1000 scanned ports on 192.168.222.130 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:0C:29:36:81:5D (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 27.17 seconds
```

How it works:

- Sends probes to discovered open ports.
- Analyses service responses and protocol banners.
- Attempts to identify the application name and version.

Security Relevance: Service version detection helps identify outdated software, unnecessary exposed services, and potentially vulnerable applications.

4.4. Operating System Detection (-O)

Purpose:

The -O option attempts to identify the operating system running on the target machine.

Command for Ubuntu:

sudo nmap -O <UBUNTU_IP> -oN ubuntu-O.txt -oX ubuntu-O.xml

Command for Windows:

sudo nmap -O <WINDOWS_IP> -oN windows-O.txt -oX windows-O.xml

```
┌──(rashmi㉿Kali)-[~/Desktop]
└─$ sudo nmap -O 192.168.222.130 -oN windows-O.txt -oX windows-O.xml
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-29 16:57 +0530
Nmap scan report for 192.168.222.130
Host is up (0.0017s latency).
All 1000 scanned ports on 192.168.222.130 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:0C:29:36:81:5D (VMware)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 28.38 seconds
```

```
zsh: corrupt history file /home/rashmi/.zsh_history
┌──(rashmi㉿Kali)-[~/Desktop]
└─$ sudo nmap -O 192.168.222.128 -oN ubuntu-O.txt -oX ubuntu-O.xml
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-29 15:37 +0530
Nmap scan report for 192.168.222.128
Host is up (0.0015s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
80/tcp open  http
MAC Address: 00:0C:29:AB:09:54 (VMware)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.19
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 6.20 seconds
```

How it works:

- Sends specially crafted TCP, UDP, and ICMP probes.
- Analyses the responses received from the target.
- Compares TCP/IP stack characteristics against Nmap's OS fingerprint database.

Important Note:
OS detection works best when Nmap can identify at least one open and one closed TCP port. Firewalls and network configurations can affect accuracy.

Security Relevance: OS detection helps security analysts understand the operating systems present in a network and identify potentially outdated or unsupported systems.

4.5. Aggressive Scan (-A)

Purpose: The -A option enables multiple advanced Nmap detection features.

Command for Ubuntu:

sudo nmap -A <UBUNTU_IP> -oN ubuntu-A.txt -oX ubuntu-A.xml

Command for Windows:

sudo nmap -A <WINDOWS_IP> -oN windows-A.txt -oX windows-A.xml

```
┌──(rashmi㉿Kali)-[~/Desktop]
└─$ sudo nmap -A 192.168.222.128 -oN ubuntu-A.txt -oX ubuntu-A.xml
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-29 15:38 +0530
Nmap scan report for 192.168.222.128
Host is up (0.0011s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
80/tcp open  http    Apache httpd 2.4.58 ((Ubuntu))
|_http-title: Apache2 Ubuntu Default Page: It works
|_http-server-header: Apache/2.4.58 (Ubuntu)
MAC Address: 00:0C:29:AB:09:54 (VMware)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.19
Network Distance: 1 hop

TRACEROUTE
HOP RTT     ADDRESS
1   1.05 ms 192.168.222.128

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.68 seconds
```

```
┌──(rashmi㉿Kali)-[~/Desktop]
└─$ sudo nmap -A 192.168.222.130 -oN windows-A.txt -oX windows-A.xml
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-29 17:01 +0530
Nmap scan report for 192.168.222.130
Host is up (0.00044s latency).
All 1000 scanned ports on 192.168.222.130 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:0C:29:36:81:5D (VMware)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

TRACEROUTE
HOP RTT     ADDRESS
1   0.44 ms 192.168.222.130

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 33.60 seconds
```

How it works:

Aggressive scanning enables:

- OS detection
- Service version detection
- Default NSE script scanning
- Traceroute

Security Relevance: Aggressive scanning provides a broader view of a target system's exposed services and possible security weaknesses.

Important: This scan generates more traffic and may be more intrusive than basic scans. Use it only against your own authorised lab machines.

5. Scan Output Files

All scans are saved in two formats.

Option	File Format	Purpose
-oN	Normal Output (.txt)	Human-readable scan results
-oX	XML Output (.xml)	Structured output for tools and further analysis

Example:

sudo nmap -sS <UBUNTU_IP> -oN ubuntu-sS.txt -oX ubuntu-sS.xml

sudo nmap -sS <WINDOW_IP> -oN windows-sS.txt -oX windows-sS.xml

The files can be reviewed using:

cat ubuntu-sS.txt

To inspect XML:

less ubuntu-sS.xml

6. Wireshark Packet Capture - TCP SYN Scan

Objective

To capture and analyse the network packets generated during an Nmap TCP SYN scan. This activity helps understand how Nmap scanning activity can be identified through packet analysis.

Step 1: Open Wireshark

- Launch Wireshark on Kali Linux.
- Select the network interface connected to the virtual machine network.
- Start packet capture before executing the Nmap scan.

Step 2: Execute SYN Scan

Open another terminal and execute:

sudo nmap -sS <UBUNTU_IP>

You can also repeat the process against the Windows VM:

sudo nmap -sS <WINDOWS_IP>

```
┌──(rashmi㉿Kali)-[~]
└─$ sudo nmap -sS -p 1-1000 192.168.222.128
[sudo] password for rashmi:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-29 16:27 +0530
Nmap scan report for 192.168.222.128
Host is up (0.0055s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
80/tcp open  http
MAC Address: 00:0C:29:AB:09:54 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 4.81 seconds
```

```
┌──(rashmi㉿Kali)-[~/Desktop]
└─$ sudo nmap -sS -p 1-1000 192.168.222.130
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-29 17:04 +0530
Nmap scan report for 192.168.222.130
Host is up (0.00048s latency).
All 1000 scanned ports on 192.168.222.130 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:0C:29:36:81:5D (VMware)

Nmap done: 1 IP address (1 host up) scanned in 25.76 seconds
```

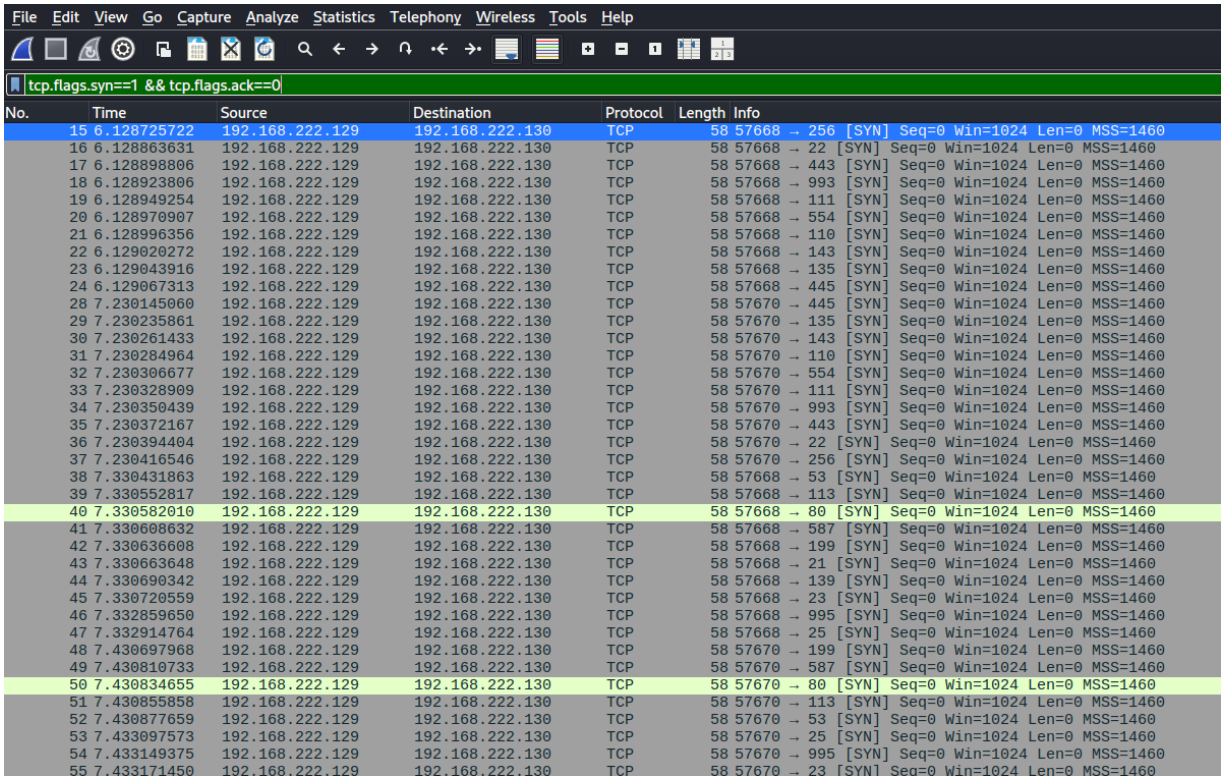
Step 3: Apply Wireshark Display Filter

Use the following display filter:

tcp.flags.syn==1 && tcp.flags.ack==0

Filter Explanation:

- tcp.flags.syn==1 identifies TCP packets with the SYN flag enabled.
- tcp.flags.ack==0 excludes packets with the ACK flag enabled.
- Together, the filter identifies initial TCP SYN packets, including connection attempts made during a SYN scan.



Step 4: Analyse Packet Responses

Observe the following packet patterns:

Packet	**Meaning**
SYN	Nmap initiates a connection attempt
SYN-ACK	Target indicates that the port is open
RST	Target indicates that the port is closed, or the connection is reset
No Response	Port may be filtered or packets may be dropped

Step 5: Save Packet Capture

After the scan finishes:

- Stop Wireshark capture.
- Go to File → Save As.
- Save file as: Ubuntu-SYN-Scan_result.pcap & Windows-SYN-Scan_result.pcap

7. Security Analysis

From a SOC analyst's perspective, Nmap scanning activity is important because attackers frequently perform reconnaissance before attempting exploitation.

Important indicators include:

- Multiple TCP SYN packets sent from one source IP.
- Attempts to connect to multiple destination ports.
- Repeated connection attempts within a short time.
- SYN packets followed by SYN-ACK or RST responses.
- Scanning activity targeting several hosts in the same network.

A SOC analyst can use these indicators to investigate possible port scanning activity.
Connection to Week 3 - IDS and Snort

The TCP SYN scan performed in this practical is relevant to the next week's IDS activities. An IDS such as Snort can analyse network traffic and generate alerts when suspicious scanning patterns are detected.

For example, repeated SYN packets sent to different ports may indicate reconnaissance activity.

The Wireshark filter used in this practical:

```
tcp.flags.syn == 1 && tcp.flags.ack == 0
```

helps identify initial TCP connection attempts and provides a foundation for understanding port scan detection rules.

8. Learning Outcomes

After completing this practical, I learned:

- How to discover active hosts using Nmap.
- How TCP SYN scans identify open and closed ports.
- How to detect service names and versions.
- How Nmap performs operating system fingerprinting.
- How aggressive scans combine multiple detection techniques.
- How to save Nmap scan results in normal and XML formats.
- How to capture Nmap scanning activity using Wireshark.
- How to identify SYN, SYN-ACK and RST packets.
- How port scanning patterns can help SOC analysts identify reconnaissance activity.

9. Conclusion

In this practical, I explored five important Nmap scanning techniques against my Ubuntu and Windows virtual machines.

I used host discovery, TCP SYN scanning, service version detection, operating system detection, and aggressive scanning to understand the information that can be collected during network reconnaissance.

I also captured TCP SYN scan traffic using Wireshark and analysed the packet patterns generated by Nmap.

This practical improved my understanding of network reconnaissance and packet-level analysis. It also provided a foundation for detecting suspicious scanning activity using IDS tools such as Snort in Week 3.